Effective Use of AI Applications to Enhance Saudi EFL Learners' Writing Ability: A Study at the Tertiary Level

Abstract

The increasing application of artificial intelligence (AI) in education provides new possibilities to facilitate second language (L2) writing development. Nevertheless, limited research has investigated the potential of AI-driven writing aids to expand the writing performance of Saudi EFL learners at the tertiary level. This research analyzes the effects of AI tools, Wordtune and Grammarly, on academic writing among Saudi university students and examines how these tools can be deliberately integrated as part of structured writing instruction. Using quantitative data, 62 students participated, with a group in each gender category assigned as an Experimental Group (EG) using AI tools and a Control Group (CG) following old methods. Descriptive Statistics, Independent-samples t-tests, and Regression Analysis in SPSS were conducted to compare pre- and post-writing tests, assessing changes in content development, coherence, cohesion, and grammatical accuracy. The EG achieved higher scores than the CG, with improvements in content development (57.40-75.20), coherence (55.60-74.10), cohesion (54.80-73.20), and grammatical accuracy (53.90-72.50). Independent t-tests showed extremely significant differences ($p < 0.001$) with large effect sizes (d=1.31-1.34), and regression analyses confirmed AI exposure as a strong predictor of writing performance (R The effects of AI were

consistent across genders. Overall, the result indicates that pedagogically integrated AI writing tools can effectively enhance Saudi EFL students' academic writing, support their independence, confidence, and overall writing competence while overcome limitations of conventional instruction.